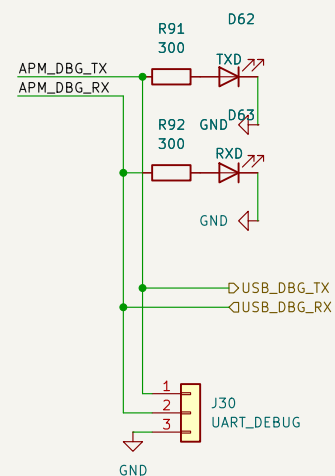


UART -> USB Debug Port

UART debug messages available via a USB port and a 3 pin jumper for TX, RX, and GND UART connections

USB2.0 DBG PORT



Sheet: /USB_debug_APM/
File: USB_debug_APM.kicad_sch

Title:

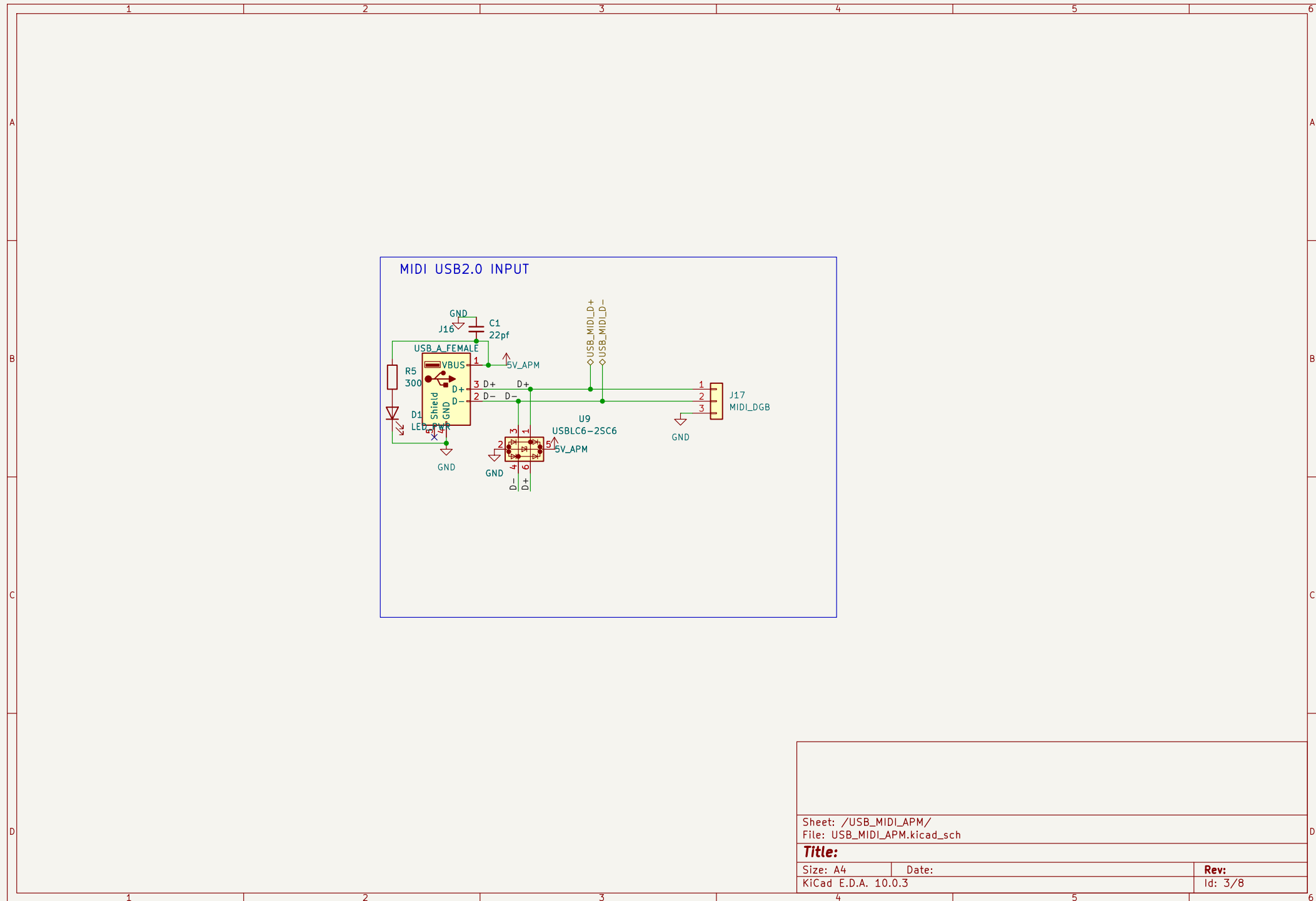
Size: A4

Date:

KiCad E.D.A. 10.0.3

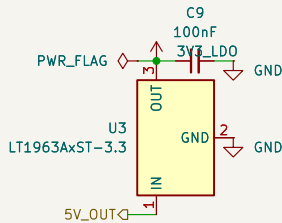
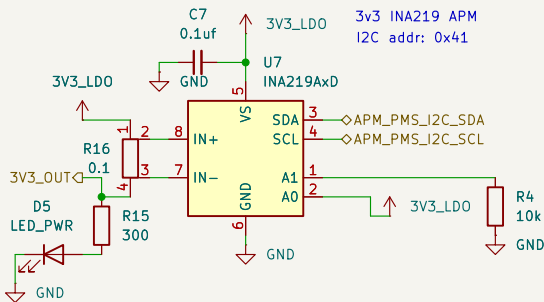
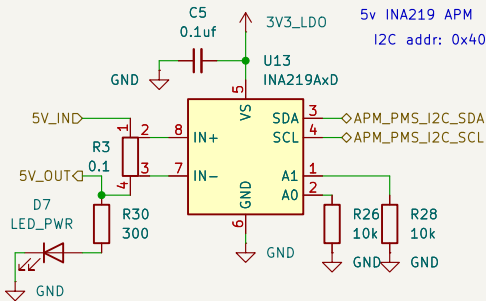
Rev:

Id: 2/8



Power Monitor and Supply

PMS has an INA219 current sensor
for each board/node in the
PCB. Each takes in the 5V from it's
parent board and supplies 5V to said
board's circuit (its peripherals)
thus enabling monitoring



Sheet: /PMS_APM/
File: PMS_APM.kicad_sch

Title:

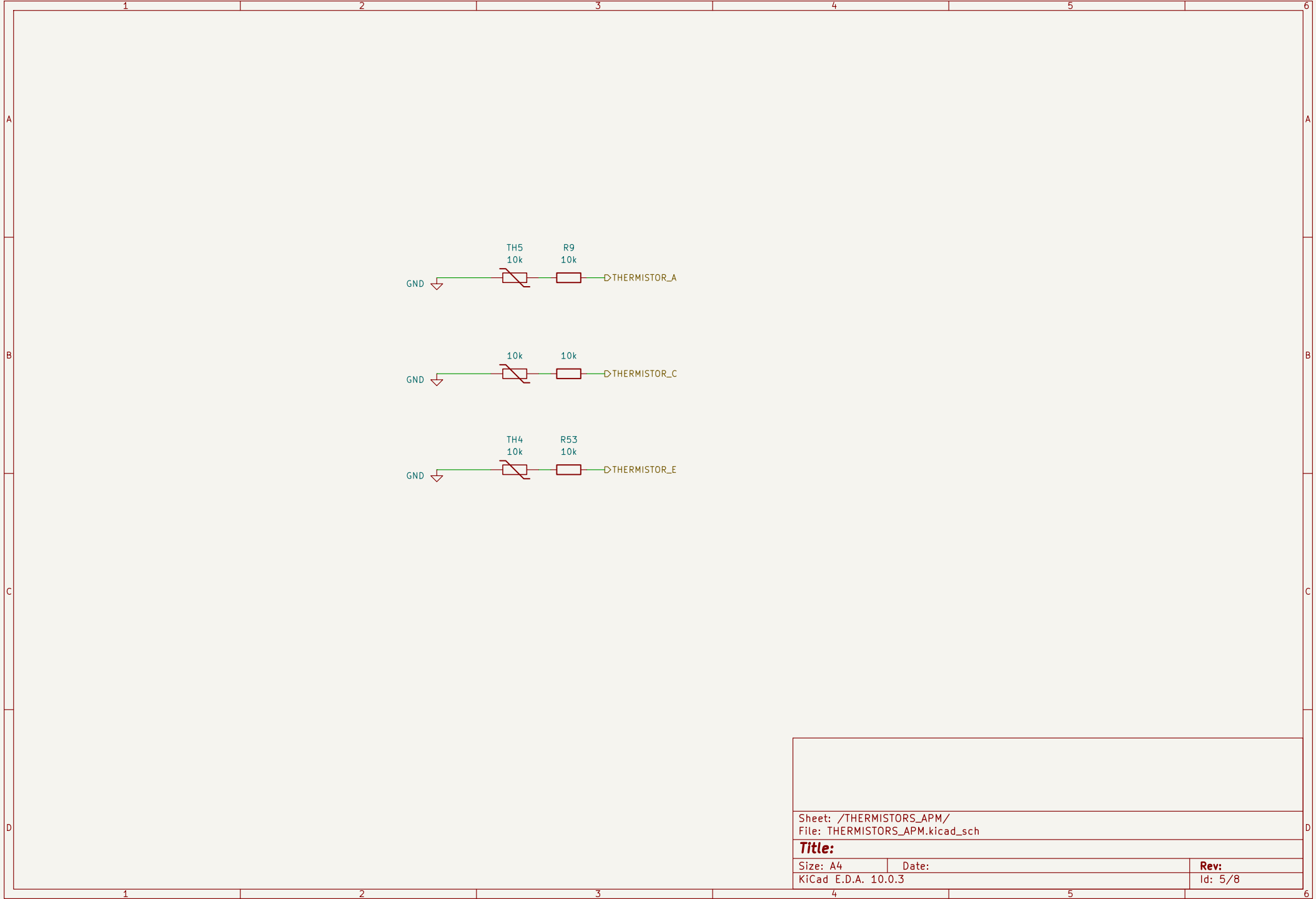
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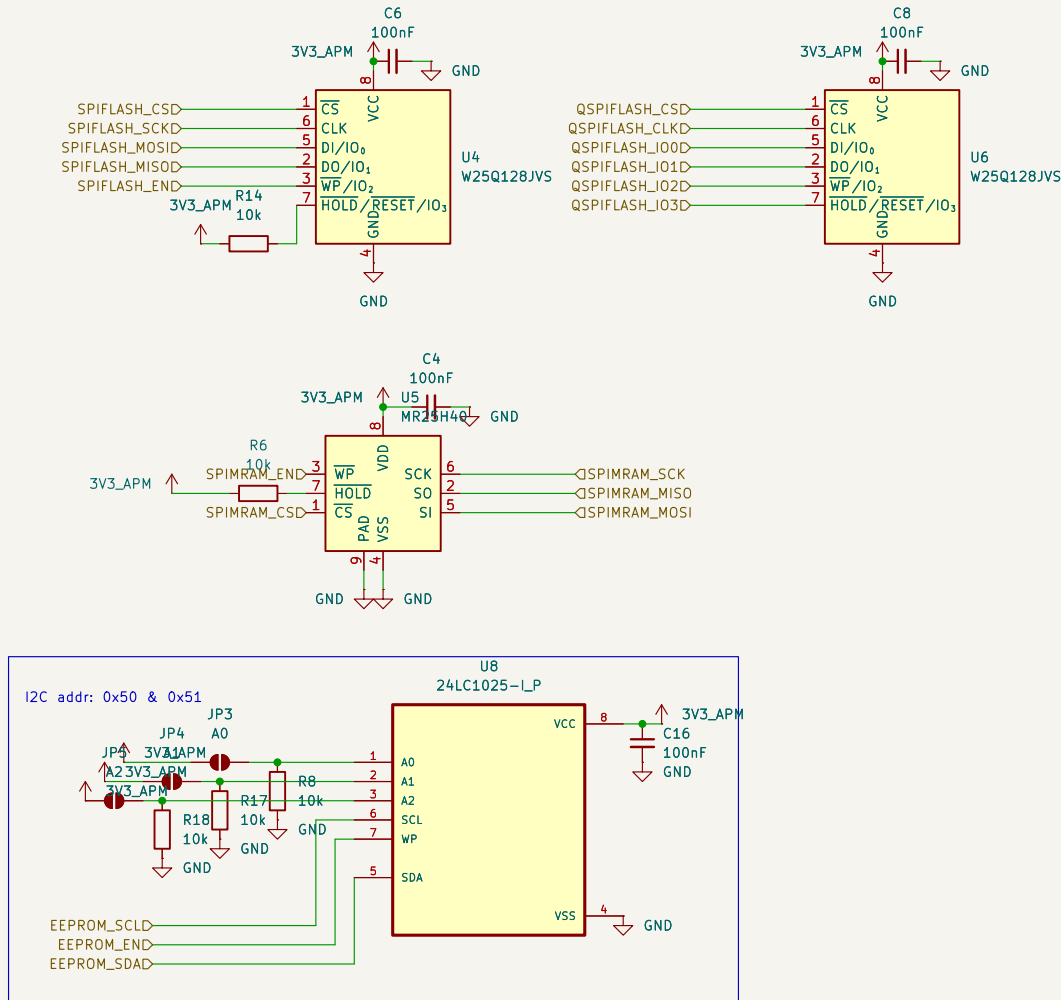
Date:

KiCad E.D.A. 10.0.3

Rev:

Id: 4/8





Sheet: /STORAGE_APM/
File: STORAGE_APM.kicad_sch

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KiCad E.D.A. 10.0.3

Id: 6/8

ACM FPGA Interface

Dual FPC STM32<->Tang Nano 20K FPGA connection configuration. The STM32 side has two connectors where only one can be active at a time

- Configuration 1: FPGA_FFC with SPI, QSPI, UART connections.
- Configuration 2: FPGA_FMC with:
 - 16 bit muxed data
 - and async FMCconfiguration resulting in a 64MB shared address space between the STM32 & FPGA

